Conditional Matrix Flows for Gaussian Graphical Models [1]

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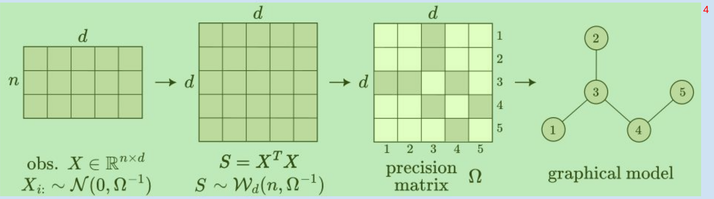
NEURAL INFORMATION PROCESSING SYSTEMS

Gaussian Graphical Models: Graphical Lasso [3]



[4]

- GGMs provides **structural** information: $\Omega_{ij} = 0$ implies conditional independence of features i and j given all remaining features [5]
- Maximum likelihood estimate of Ω is not sparse and for $d > n$ is not unique → **penalized likelihood** to encourage sparse Ω

$$\text{argmax}_{\Omega \succ 0}\{\log\det\Omega - \text{Tr}(\tfrac{1}{n}S\Omega) - \lambda\|\Omega\|_0\}$$

Intractable because the objective is highly non-convex [6]

- **Graphical Lasso:** use convex relaxation $\|\Omega\|_0 \longrightarrow \|\Omega\|_1$ [7]

Benefits [8]

❖ **Solution path**: frequentist approach provides solutions as a function of λ [9]

Limitations [10]

❖ **Over-shrinking:** l_1 norm causes well-known over-shrinking of features [11]
❖ **No credibility interval**

Bayesian Graphical Lasso [12]

- Bayesian approach: $p(\Omega|S,\lambda) \propto p(S|\Omega)\ p(\Omega|\lambda)$ [13]
- Encourage sparsity with double exponential prior (DE)

$$p(S|\Omega) \propto \mathcal{W}_d(n, \Omega^{-1})$$ [14]

$$p(\Omega|\lambda) \propto \mathcal{W}_d(d+1, \lambda 1_d) \prod_{i<j} \text{DE}(\omega_{ij}|\lambda) I[\Omega \succ 0]$$ [15]

- Recovers the l_1 norm penalized likelihood solution in the MAP limit [16]

Benefits [17]

❖ **Credibility interval:** can be computed within Bayesian framework [18]

Limitations [19]

❖ **Expensive posterior sampling:** needs Gibbs sampler (also limits prior choice) [20]
❖ **Over-shrinking:** prior cannot be generalized to sub-l_1 pseudo-norms
❖ **Expensive solution path**: independent Markov chains for each λ value

Conditional Matrix Flow for Gaussian Graphical Models [21]

Goal: **unify benefits** of frequentist and Bayesian frameworks while **overcoming limitations** [22]

We propose a general framework for **variational inference** with matrix-variate Normalizing Flows in GGMs. As a key improvement on previous work, we train with one flow a continuum of sparse regression models jointly for all regularization parameters λ and all l_q norms, including non-convex **sub-l_1 pseudo-norms**. [23]

PROPOSED APPROACH [24]

VARIATIONAL INFERENCE [25]

- Posterior approximation and efficient sampling via **Normalizing Flow**: $\text{D}_{\text{KL}}\left(\hat{p}(\Omega|S,\lambda)||p(\Omega|S,\lambda)\right)$ [26]
- Free choice of prior $p(\Omega|\lambda)$ and likelihood $p(S|\Omega)$ → model any Bayesian GGMs beyond Lasso

SUB-l_1 PSEUDO-NORMS [27]

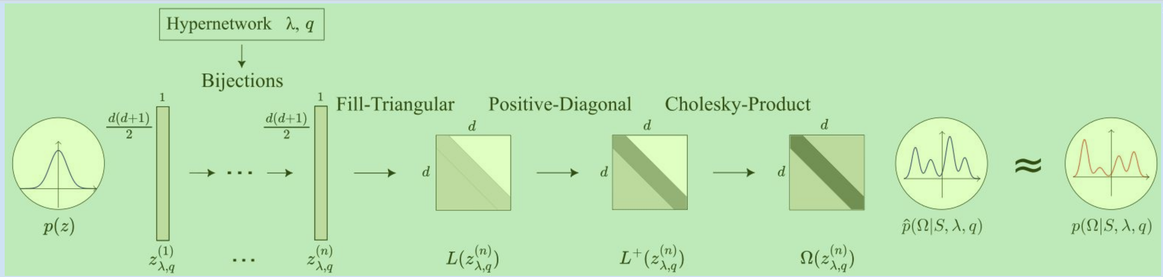
- Generalized Gaussian Distributions (GGD) as sparsity-inducing priors corresponding to **l_q (pseudo-) norms** [28]

$$\text{GGD}(x|\lambda, q) = \frac{q\lambda^{1/q}}{2\Gamma(1/q)}\exp\{-\lambda|x|^q\}$$ [29]

CONDITIONAL FLOW [30]

- Condition the Normalizing Flow on λ and q to obtain **solution paths** [31]
- Can select λ by maximizing marginal likelihood

ARCHITECTURE [32]



[33]

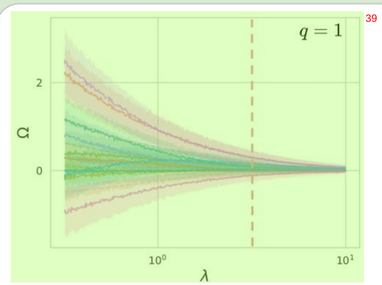
- Condition bijections on λ and q and define flow on space of symmetric positive definite matrices **by construction** [34]
- **Efficient log determinant** of Jacobian for Cholesky-Product → $\det \mathcal{J}_{\text{Chol}}(L) = 2^d \prod_{i=1}^{d} (L_{ii})^{d-i+1}$

Combine benefits [35]

❖ **Credibility intervals:** sample-based credibility intervals for any λ and q [36]
❖ **Solution paths**: we recover Bayesian Graphical Lasso ($T = 1$) and Graphical Lasso solution paths ($T \to 0$) by training with simulated annealing

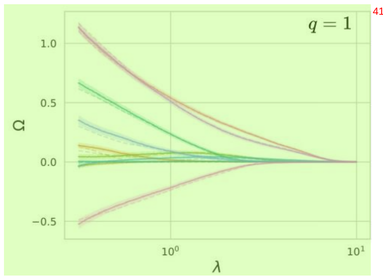
Overcome limitations [37]

❖ **Cheap posterior sampling**: no need for sequential Gibbs sampler [38]
❖ **No over-shrinking:** GGD prior enables exploration of sub-l_1 pseudo-norms
❖ **Model selection**: direct access to the marginal likelihood for selecting λ



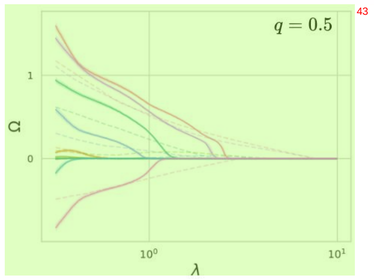
[39]

Bayesian solution path at $T = 1$ [40] **& model selection**



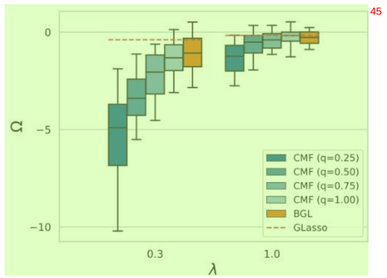
[41]

Frequentist solution path at $T \to 0$ [42]



[43]

Solution path in **sub-l_1** regime [44]



[45]

Credibility intervals in sub-l_1 regime: [46] **no over-shrinking**

[47]

References [48]

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